



YOUTH AI LAB COMPREHENSIVE APPROACH

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The Youth AI Lab programme, its ambitions, and what makes it different from a technology course.



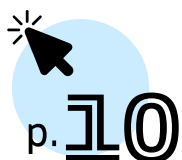
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Introduction

A scientific investigation laboratory for young people, rooted in social reality

Funded by the European Union through the **Erasmus+ programme as part of the Youth cooperation initiatives**, the Youth AI Lab project aims at creating a **documented, tested and replicable laboratory creation methodology**, giving young people the opportunity to explore the **functioning of generative AI**.



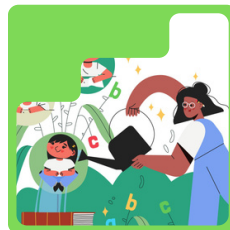
Demystify AI through
hands-on
experimentation



Develop critical thinking
and literacy about
technology



Empower youth as
creators, not just
consumers



Build transferable
resources for youth
educators

AI tools are becoming increasingly popular and accessible, yet a significant part of the youth population remains at a distance from digital technologies, exposed to their effects without the means to understand or navigate them. This digital divide is not a minor footnote: it shapes who gets to question these tools and who simply undergoes them. Understanding a technology means giving yourself the means to question it, refuse it, or transform it.

The core premise of this programme is clear: **young people in the Youth AI Labs are not consumers of AI**. They are investigators, formulating hypotheses, validating them, and transferring their findings as open research material, opening the black boxes related to algorithms, biases, usages and societal impacts. By proposing a genuine scientific process in the field of informal education and youth initiatives, we want to demonstrate the relevance of a researcher's approach to structuring the engagement of young people in their out-of-school activities.



This programme does not separate scientific rigour from socio-cultural relevance. **Scientific mediation and socio-cultural facilitation are not opposites: they reinforce each other.**

The facilitator's role here goes beyond introducing a technology. Youth workers, educators and community animators occupy a **unique position in young people's lives**: they accompany them at a moment when identities are being formed, when futures are being imagined, when choices about who to become are, often quietly, being made. Bringing AI into that space is not a neutral act. It is an opportunity to help young people see themselves as **capable of understanding complex systems**, of **contributing to technical and societal debates**, of **choosing paths they might not have considered**.

This is also why the question of AI **cannot be separated from a broader vision of society**. What kind of world do these tools reflect? Who designed them, for whom, and with what values? Engaging young people with these questions is not a distraction from the technical work; it is its purpose. The Youth AI Lab is built on the conviction that **these questions belong to everyone and that young people, given the right conditions, are entirely capable of making them their own**.



Our approach

The approach unfolds across six phases that follow the logic of a genuine scientific investigation.

It opens with an act of curiosity and destabilisation (**Phase 1: Observation and Documentation - Getting inside the machine**), moves into conceptual grounding (**Phase 2: Hypothesis Formulation - Understanding the mechanics**), then into hands-on construction (**Phase 3: Experimentation - Building**).

From there, participants deploy what they have made in real conditions (**Phase 4: Practical Validation - Deploying**), before stepping back to examine it critically (**Phase 5: Discussions and Reflections - Critical distance**).

The cycle closes with an act of transmission (**Phase 6: Communication - Sharing**), because knowledge that stays within the group is knowledge half-finished.

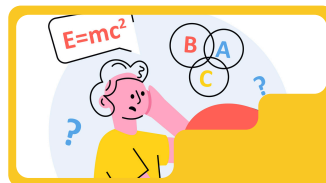
Observation

Construct a definition of generative AI



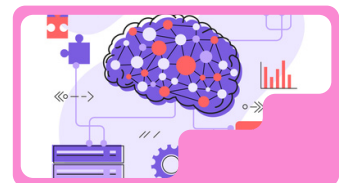
Hypothesis

Build hypothesis on how algorithms work



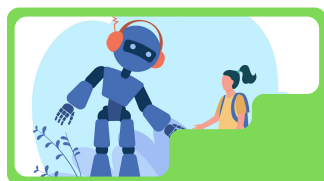
Experimentation

Create an SDG-oriented language model



Validation

Use the model for a purpose including robotics



Reflection

Debate on the learnings and usages



Communication

Share discoveries through publications



Phase 1: Observation and Documentation

Getting inside the machine: play, observe, question

Before anything else, we need to break through the fascination or suspicion that generative AI can inspire. So we start by playing with it, and above all, by trying to catch it out. Activities such as "Looks Right... Is It Right?", "Try to Break It!" or "The Stupid Invention" invite participants to actively test predictive models, document what works, what fails, and why. The goal is not technical: it is to build a critical stance from the very beginning, showing that these tools are powerful for certain tasks... and remarkably limited for others.

Themes explored

predictive AI, model capabilities and limits, critical use of digital tools

Skills developed

curiosity, observation, critical thinking, ability to ask good questions

Phase 2: Hypothesis Formulation

Understanding the mechanics: algorithms, biases and societal issues

Once curiosity is awakened, we dive into how algorithms work through unplugged, hands-on activities before any digital tool is introduced. Activities such as workshops on bio-inspired algorithms, a bias challenge to understand how biases enter language models, or an exploration of the links between AI and societal challenges (SDGs, ecological transitions, health, education) help participants build shared vocabulary and solid conceptual understanding. Robot design requirements are introduced at this stage, linking robotics and LLMs from the outset.

Themes explored

bio-inspired algorithms, bias in AI, SDGs, robotics design

Skills developed

understanding of systems, bias identification, connection to real-world issues



Phase 3: Experimentation

Building: datasets, models and compared versions

This is the core of the experimentation. Activities such as dataset collection around an SDG challenge, fine-tuning a base model, and creating multiple versions in teams to compare, discuss and optimise form the backbone of this phase. Particular attention is given to multilingual challenges: how does a model behave across different languages? What information is lost? What inequalities does this reveal? This dimension is especially valuable in an Erasmus+ programme that brings together teams from different countries.

Themes explored

data collection and preparation, language model fine-tuning, multilingual AI

Skills developed

foundational technical skills, experimental rigour, cooperation, iterative thinking

Phase 4: Practical Validation

Deploying: a speaking agent in service of citizens

The models built are turned into concrete conversational agents. Participants must shift from project mode into a use-centred design mindset: who are we talking to? What is actually useful? What is actually understandable? A social and societal validation session is organised: the group presents their agent to external users (local residents, partners, other young people) and collects genuine feedback on effectiveness and perceived usefulness.

Themes explored

conversational agents, user-centred design, societal validation

Skills developed

communication, empathy, sense of public utility, confrontation with reality



Phase 5: Discussions and Reflections

Critical distance: biases, regulation, costs and limits

Once the project is built and tested, it is time to step back collectively. This phase is deliberately reflective and sometimes uncomfortable. It invites participants to evaluate what they have produced through ethical and political lenses. Activities such as bias analysis of their own outputs, comparison of regulatory frameworks (GDPR vs US legislation, EU AI Act...), or real-life application of the robot and LLM with cost/benefit analysis open deeper questions: scalability, data ownership, energy consumption. These discussions are not optional extras at the end of the journey; they are an integral part of developing a citizen's perspective on technology.

Themes explored

algorithmic bias, AI regulation, energy and data ownership, cost-benefit analysis

Skills developed

applied ethics, critical thinking about technology, ability to debate complex issues

Phase 6: Communication

Sharing: making the work visible and reusable

Every research project ends with communication. Activities such as reaching out to local elected officials or institutions, publishing findings (academic or accessible), organising an exhibition or a competition, or building a documented LLM project library give participants real ownership over their work and extend its reach beyond the group. A major collective event, ideally connected to a Maker Faire or equivalent, which will bring all partner teams together to share their work and be recognised for what they have achieved.

Themes explored

scientific and public communication, open access, project documentation

Skills developed

communication, pride in one's work, open-source culture and knowledge sharing



Our Socio-Educative Roadmap

The Youth AI Lab is designed to be documented, tested and replicable by any youth organisation, fablab, youth centre, or informal education structure. The replicable unit is simple: a minimum of 10 workshops covering the six phases described in this document. Beyond this minimum, the format is deliberately open: implementations across the consortium already range from weekly two-hour sessions spread over several months to intensive residential formats. Each will be fully documented, including adjustments and difficulties, so that future facilitators inherit honest evidence rather than an idealised model.

The programme targets young people aged 12 to 17, a deliberately wide range. The methodology does not expect the same from a 12-year-old and a 17-year-old: each phase is designed to be accessible at entry level and deepened at higher levels, so facilitators can calibrate without changing the structure. In its full form the programme runs over one to two years, but shorter standalone formats are equally valid and will be documented as such. All resources are shared in open access.

FACILITATION APPROACHES

Educator, animators and youth workers are not expected to be AI experts: their role is to **hold the group, maintain the investigative posture, and connect the technical work to the social dimensions** that give it meaning.

In addition, the Youth AI Lab approach introduces **mentors**, typically young adults aged 18 to 25, who are not supervisors but **inspirers through proximity**: they make technical and academic paths feel accessible and real to participants who might not otherwise see themselves in them.

A few principles cut across all contexts. **The group investigates together**: facilitators ask more questions than they give answers. **Local context is a resource**: every SDG challenge, every user, every community served is real. **Inclusion is structural**: entry through play, debate and making is designed to reach young people who do not see themselves as "tech people", including girls and those from underrepresented backgrounds.

And youth are not just participants: **they can take on roles in governance, communication and evaluation throughout**.



EVALUATION TOOLS

The evaluation is structured in three moments. **Before**, participants, mentors and organisers complete baseline questionnaires on skills, expectations and motivations, a reference point for measuring change and identifying where support is needed early. **During**, evaluation stays light and embedded: closing rituals at the end of each session (exit tickets, sticky notes, short scales), a brief implementation diary kept by mentors, and the AI Maker Skill Tree used at intervals to help participants track their own progression. **After**, a post-assessment mirrors the pre-assessment; optional tools include a group interview protocol and a skills rubric covering problem framing, SDG alignment, ethics, robotics integration and collaboration.

The framework was developed by UAB and validated collectively by the consortium. All tools will be available in open access and in multiple languages, so that any replicating organisation **can use the same instruments and compare results**.

Ready for Youth Ai Lab

A methodology made to be taken, adapted, and run by others

What this approach is not

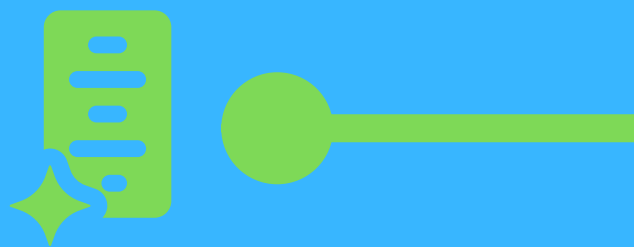
- **It is not a disguised computer science class.** Technical skills are means, not ends.
- **It is not an introduction to AI "to do what adults do".** Young people produce something real, for real uses.
- **It is not an approach reserved for top students or tech enthusiasts.** Entry through play, debate, physical making and collective challenge ensures broad accessibility.

What this approach is

- **It is a scientific investigation programme.** Young people formulate hypotheses, run experiments, and produce results that others can build on.
- **It is a making programme.** Participants leave with a language model, a robot, and a documented project they created themselves.
- **It is a citizenship programme.** Every technical choice made along the way is examined for what it says about society, power, and the kind of future we want to build.



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<https://www.youth-ai-lab.eu/>

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